Dias Azhigulov

Vancouver, British Columbia
Email: dias.azhigulov@ece.ubc.ca
LinkedIn: www.linkedin.com/in/azhigulov/
GitHub: github.com/dazhigulov
Tel: +1 (236) 668-7648

EDUCATION

University of British Columbia (Vancouver, BC, Canada) **Jan 2021 – Present**

Master of Applied Sciences in Electrical and Computer Engineering (GPA: 4.30/4.33)

- Research area is analog computing through Silicon Photonics.

Nazarbayev University (Nur-Sultan, Kazakhstan) **Aug 2016 – June 2020**

Bachelor of Engineering in Electrical and Electronic Engineering (graduated with honors, GPA: 3.73/4.00)

EXPERIENCE

Graduate Teaching Assistant **Aug 2021 - Present**

University of British Columbia – Vancouver, BC, Canada

- Delivered workshops on Linux and Git to undergraduate students.

Graduate Research Assistant **Feb 2021 – Present**

System-on-Chip Lab, University of British Columbia – Vancouver, BC, Canada

- Working on the design of a scalable logic devices based on Mach-Zehnder Modulators (MZM) and Ge photodetectors. The simulation is being performed in Cadence (through custom Verilog-A models) while the layout is being drawn in KLayout with verification done through Mentor Calibre DRC. This device is to be fabricated by AMF Singapore.
- Using simulation data from Ansys Lumerical, developing a Verilog-A model for Microring Modulators that can output accurate behavior of the device based on the user-defined design parameters (coupling length, gap, ring radius). The model will account for static and dynamic nonlinearities and provide realistic estimation for bandwidth.
- Conducted lab measurements on two Si-Photonics chips fabricated by AMF Singapore and Applied Nanotools Inc. (tested devices include slotted waveguides, ring resonators, mode converters, MZM with no metal layer). Currently preparing a setup for active device measurements.
- Designed Fabry-Perot cavities and various interferometers for fabrication and subsequently performed experimental parameter extraction (coupling coefficient, Q-factor, effective index) from the chip measurements and compared the results with simulation.

Data Scientist **Sep 2020 – Feb 2021**

"Fund for supporting research and development of artificial intelligence" CF – Nur-Sultan, Kazakhstan

- Wrote a script in Python that extracts various statistics about audio transcriptions (SNR, speech length etc.) and preprocesses them along with open-source datasets for Automatic Speech Recognition. Trained hybrid HMM-DNN models in Kaldi on the dataset of phone calls (bilingual speech) and achieved 43% WER on real world data.
- Built robust scalable address converter in Java via Apache Lucene and Regex. The final accuracy is 91% (i.e. the program can match 91 out of every 100 raw addresses), the time it takes to find matches from CSV to 10000 inputs is 1 hour. The matched addresses (text data) are then used to obtain their approximate geocoordinates via ArcGIS engine. The entire pipeline was realized in Ubuntu.

Undergraduate Research Assistant **Oct 2019 – May 2020**

Data Analytics Lab, Nazarbayev University – Nur-Sultan, Kazakhstan

- Developed a Machine Learning based web application which can in live mode detect different cancers and determine the type of cancers with an accuracy of over 99% based on the images uploaded by a user. Frontend is implemented in Python Flask, Backend is built via Keras and consists of 9 CNNs trained on open-source clinical data. For mobile application purposes the models were also compressed up to 2 times without loss in accuracy through channel pruning technique.

- Using Python created a website that visualizes the risk of getting COVID-19 in major cities of Kazakhstan (https://covid-19-kz.herokuapp.com). The data is automatically updated each day.
- Built a pipeline in Python for preprocessing and classifying human brain signals with Transfer Learning as part of Brain-Computer Interface project (72% accuracy, subject-independent testing). A range of CNNs (VGGNets, Xception, DenseNets) were studied.

Undergraduate Research Assistant **Sep 2018 – May 2020**

iDSN Lab, Nazarbayev University – Nur-Sultan, Kazakhstan

- Designed seven basic Si-Photonics logic gates in Ansys Lumerical through microring resonators and heaters. Results are published in several SPIE conferences and in an academic journal.
- Developed a script in Python for extracting statistical data from 5 torrent tracker websites through HTML parsing. The data was used to analyze the Central Asian BitTorrent network and findings were presented at the CoCoNet 2018 conference.

Visiting Student **May – Aug 2019**

ICS Lab, KAUST – Thuwal, Kingdom of Saudi Arabia

- Designed several NEM relay-based power converters (DC-DC, AC-DC, and DC-AC) in Cadence Virtuoso. Results are published in the ISCAS 2020 conference proceeding.

COURSE PROJECTS

- Had experience designing a variable frequency divider for PLLs with custom logic to switch between different frequency bands, clock signal distribution systems (56 ps skew, 375 ps latency, 65 mW power consumption), and 256x256 array of 6T SRAM cells (schematic & layout) in Cadence Virtuoso.
- Designed error amplifiers (simple differential pair and cascode configuration) for LDO using LTSpice.
- Deployed a real-time face recognition system on Raspberry Pi 3 for automated attendance monitoring. Face detection and tracking was implemented in Python through OpenCV, and the classification results were programmed to be uploaded to the AWS IoT server via the MQTT protocol.
- Using Arduino and ESP 8266 module built a UART protocol based communication link to exchange messages between two computers (emulation of client-server interaction).
- Devised an algorithm and testbench in VHDL to solve maze escape problem by sensing surroundings.
- Implemented a fire-alarm circuit on PCB using thermistor and buzzer.

PUBLICATIONS

- *BCI 2021 (South Korea)*: "Deep Transfer Learning for Subject-Independent ERP-based BCIs". *ISCAS 2020 (Seville)*: "BEOL NEM Relay-Based Inductorless DC-DC Converters". *Journal of Optical and Quantum Electronics*: "High-speed thermally tuned electro-optical logic gates based on micro-ring resonators". *SPIE Photonics West 2020 (San Francisco)*: "Design of Si-photonics-based logic gates using micro-ring resonator structures". *SPIE Photonics West 2019 (San Francisco)*: "The design of universal logic gates using micro-ring resonator structures". *CoCoNet 2018 (Nur-Sultan)*: "Analyzing the BitTorrent Ecosystem of Central Asia".

TECHNICAL SKILLS

- Cadence Virtuoso; LTSpice; Lumerical; KLayout; Mentor Graphics Calibre; Coventor MEMS+; Quartus Prime; Verilog-A; Verilog; VHDL; Python (Tensorflow, PyTorch, OpenCV, Numpy, Pandas and other ML libraries); Python Flask; MATLAB; SQL; C; Embedded C; C++; Java; Git; Linux; Docker; HTML & CSS.

AWARDS

- DAAD Scholarship winner (22000 EUR worth of funding). **Apr 2020**
- 3000 USD research grant from the Young Researchers Alliance (yrascience.kz). **Dec 2019**
- Invited to attend KAIST EE Camp (best university in South Korea). **Aug 2019**
- Certificate of Appreciation for volunteering in the IEEE Student Branch **Apr 2019**

LANGUAGES

- Kazakh – Native; Russian – Fluent; English – Fluent (IELTS: 8.0); German – Intermediate (B1)